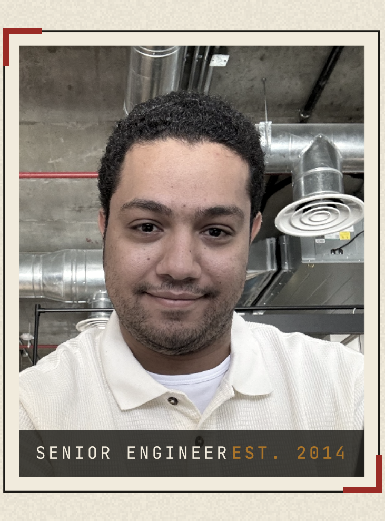
NO. 18 · VOL. ENGINEERING CURRICULUM VITÆ — THE ENGINEER'S RECORD EST. MMXIV

Ahmad Abdul-Jawad *Mahmoud*



SENIOR SOFTWARE ENGINEER / IOS · ANDROID · BACKEND

CAIRO → RIYADH · 12+ YEARS SHIPPING PRODUCTION SOFTWARE

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● SEE THE LIVE INTERACTIVE EDITION ↗

Senior software engineer with **12+ years** shipping production software — native iOS first, now across Android, backend and web. I have built apps used by 35M+ people: HungerStation, Ego ride-hailing, and for seven years the Marn POS used by 20k+ restaurants and cafés, where I owned an offline-first CoreData core with cloud sync and held a **99.8–99.9% crash-free** rate. Right now I am building a production ride-hailing platform end to end, **solo**: a NestJS / PostgreSQL / Redis backend, a Next.js operations console, and native iOS & Android apps on one API contract.

12+ YEARS HANDS-ON

35M+ USERS REACHED ACROSS SHIPPED APPS

99.8%+ CRASH-FREE RATE IN PRODUCTION

3,700+ AUTOMATED TESTS ON MY CURRENT PLATFORM

§ 01 How I Engineer

01 — SYSTEMS DESIGN

Design for the failure case

An offline-first CoreData + cloud-sync core so a venue keeps trading when the network does not. A **trip state machine**, **scored dispatch** over a Redis geo index and a **double-entry ledger** with idempotent mutations, so a retry can never double-charge.

02 — QUALITY YOU CAN MEASURE

Tests gate the merge

Automated tests that block bad merges, CI/CD built from nothing (**60% faster deployments**), and a **99.8–99.9% crash-free rate** in production. On my current platform: **2,900+ backend** and **800+ iOS** tests.

03 — SHIP WITHOUT STOPPING

Modernise in flight

A multi-year move to SwiftUI, Combine and async/await without pausing feature delivery, and a performance programme that cut load times **40%**. Swift 6 strict concurrency, actors and structured concurrency in new code.

04 — END-TO-END OWNERSHIP

Credible at every layer

Native iOS (SwiftUI) and Android (Kotlin / Compose) clients on shared, versioned core libraries; NestJS / PostgreSQL / Redis backends; a Next.js console. I build and run **all four layers of a production platform myself**.

§ 02 Professional Experience

Aug 2019 — Aug 2026
RIYADH, KSA

iOS Team Lead — 7 years leading the iOS org

Marn — Jahez Group ↗

- **Owned the architecture** of an enterprise POS used by 20k+ restaurants and cafés: an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Designed the core systems** — table management, order processing and inventory — at the heart of the product.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery.
- **Held a 99.8–99.9% crash-free rate** in production through clear module boundaries, code review and automated tests that gate merges.
- **Built CI/CD and the release process from scratch**, cutting deployment time 60%, and ran the performance programme behind 40% faster load times.
- Led and mentored a **5+ engineer iOS team** for seven years, and partnered with product, design and backend on scope and estimates.

Mar 2026 — Present
CURRENT
INDEPENDENT

Founding Platform Architect — iOS · Android · Backend · Web

Ride-Hailing Platform · Independent Project

Took the **full technical scope of a founding CTO** and carried it alone: chose the stack, set the standards, and have been building a production ride-hailing platform — a hardened backend and a real-time ops console **in production**, with native **iOS/SwiftUI** and **Android/Compose** apps (Rider & Driver on each) in active development. Every layer from data model to release, governed by explicit rules: idempotent mutations, double-entry accounting, acyclic module boundaries, and a CI test suite that gates the merge.

Backend APIs
NestJS · TypeScript · Mar '26 → now

Modular monolith, **23 modules** — **trip state machine**, **scored dispatch** over a Redis geo index, **double-entry ledger**, realtime rider/driver/admin **sockets**, JWT + phone-OTP, idempotency & audit log · ~320 endpoints over 47 controllers, traffic-aware fare quotes, **2,900+ tests**, Swagger docs.

Admin Console
Next.js 14 · Jun '26 → now

Bespoke **30-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC onboarding review, finance reconciliation, refunds/payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO**.

iOS Apps · Rider & Driver
SwiftUI · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** Swift package (domain · networking · design system, pinned remotely) — live GPS **trip lifecycle**, Socket.IO feeds with fresh-token reconnect, in-app chat, payments, Live Activities, driver earnings & turn-by-turn navigation · Swift 6 strict concurrency, **800+ tests**.

Android Apps · Rider & Driver
Kotlin · Jetpack Compose · Sep '26 → now

Native **Kotlin / Compose** ports of both apps against **the same API contract**, on a shared **buraq-core-android** library published to GitHub Packages — driver KYC onboarding, offers and trips; rider booking, live tracking, FCM push and Google Pay · **290+ test suites**.

One backend · one contract · four native clients in flight — plus growth loops (promotions, referrals, loyalty tiers, wallet), driver safety/SOS, and scheduled reconciliation jobs. The exercise a technical leader needs periodically: own every trade-off yourself, then know exactly what you are asking of a team.

Dec 2025 — Jun 2026
VOLUNTEER

Senior iOS Engineer

HaramBlur ↗ · Voluntary Charity Project

- Designed an architecture integrating on-device machine learning for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

Mar 2018 — May 2019
RIYADH, KSA

iOS Team Lead

Ego — The Ride Hailing App ↗

- Led iOS delivery for a complete ride-hailing ecosystem (rider, driver, watchOS) serving 15M+ users in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Sep 2016 — Mar 2018
KSA

Senior iOS Engineer

HungerStation — DeliveryHero Group ↗

- Contributed to a major food-delivery platform serving 20M+ users.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

May 2016 — Aug 2016

iOS Developer

ALNapedh ↗

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

Jul 2015 — Sep 2015

iOS & Apple Watch Developer

Nile Design Co. ↗

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

May 2014 — May 2015

iOS Developer

UK Pro. Solutions Ltd. ↗

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

§ 03 Selected Projects

01 — FULL-STACK

Ride-Hailing Platform (built from scratch · since Mar 2026)

SwiftUI · Kotlin · Jetpack Compose · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing platform architected solo: a hardened NestJS backend (trip state machine, scored dispatch, double-entry ledger, realtime sockets, 2,900+ tests) and a real-time Next.js operations console **in production**, with **four native clients** (Rider & Driver on iOS/SwiftUI and on Android/Kotlin·Compose) in active development.

◆ Four Platforms · Solo Ownership · Production Security

02 — ENTERPRISE

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise point-of-sale serving 20k+ restaurants and cafés across Saudi Arabia. Offline-first architecture, complex table management, real-time order processing — **the product my iOS team owned for seven years**.

★ 4.8 / 5 — 2,437 ratings

03 — MOBILITY

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · Push · watchOS

Complete ride-hailing ecosystem: rider app, driver app, and Apple Watch companion. Real-time GPS, Apple Pay, and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

04 — DELIVERY

HungerStation

UIKit · Firebase · Real-time APIs · Perf Optimization

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Focused on performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

05 — AI / PRIVACY

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur effects, and live video monitoring.

◆ AI-Powered Privacy Browser

06 — COMMUNITY

Daleelo

MapKit · Core Location · Social SDKs · Community

Muslim community app for users across the US — prayer times, mosque finder, and community features with enhanced location and social integration.

★ 4.5 / 5 — 1,287 ratings

§ 04 Stack & Craft

ENGINEERING PRACTICE — CRAFT

System Design · Automated Testing · API Design · Offline-first & Sync · Realtime (WebSockets) · Performance Profiling · CI/CD · Code Review · Modular Codebases · Versioned SDKs · Mentoring · Agile / Scrum

IOS — 2025 — CORE

Swift 6.0 · SwiftUI · UIKit · Combine · async/await · SwiftData · iOS 18+

ANDROID — 2025 — NATIVE

Kotlin · Jetpack Compose · Coroutines · Flow · MVVM

ARCHITECTURE — PATTERNS

MVVM · VIPER · Clean Architecture · Coordinator · TDD

APPLE ECOSYSTEM — PLATFORMS

watchOS · tvOS · macOS · WidgetKit · Live Activities · App Intents

MODERN FEATURES — INTELLIGENCE

Core ML · Vision · ARKit · Dynamic Island · App Tracking

BACKEND & APIS — FULL-STACK

NestJS · TypeScript · PostgreSQL · TypeORM · Redis · JWT/OTP Auth · REST · GraphQL · Firebase · CloudKit · WebSockets · Swagger/OpenAPI

TOOLING & AI — WORKFLOW

Xcode 15+ · Fastlane · TestFlight · Instruments · Git · Copilot · Cursor

§ 05 Portfolio Index

FOOD & GAMING

HungerStation — delivery
Punters — racing tips

TRANSPORTATION

Ego — Rider
Ego — Driver
Ego — Watch app

ENTERPRISE

Marn POS
Medical Guide

SOCIAL & COMMUNITY

Daleelo
NU-inc
VeeU

WATCHOS

Ego Watch
TaQs Weather
TweetMe

E-COMMERCE

iBss
Exclusive Luxe
Scratch

RELIGIOUS & EDU

Afasy
ArabNileGroup

UTILITY & TOOLS

Unlocking Smart
Insta
iContacts
🚀 *First iOS app*

AI & ML

HaramBlur — CoreML
RecognizeMe
🎓 *Grad project — OpenCV*

WEB

FaceDemo
Web Pages

§ 06 Record

EDUCATION

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

CERTIFICATION

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

LANGUAGES

Arabic	Native
English	Professional

§ 07 Honours

01 Owned the architecture of a POS platform serving **20k+ restaurants** and cafés for **7 consecutive years** — through its growth from launch scale to enterprise.

02 Held a **99.8–99.9% crash-free rate** and cut deployment time **60%** by building the standards, tests and CI/CD the team ran on.

03 Architecting a **full-stack ride-hailing platform solo** — NestJS/PostgreSQL backend and Next.js ops console in production, four native clients (rider and driver, iOS and Android) in development, with idempotent APIs and double-entry accounting.

04 Built a food-delivery platform serving **20M+ users** across Saudi Arabia and Bahrain, and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** published on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) and early SwiftUI adopter (WWDC 2019) — a habit of betting on the right platform shift early.

SENIOR SOFTWARE ENGINEER · 12+ YEARS HANDS-ON · IOS · ANDROID · BACKEND

SET IN FRAUNCES & JETBRAINS MONO
EDITION 2026 · CAIRO → RIYADH